LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

# Lesson 4: Gravity: What Goes Up...

Module 2 | 45 minutes | Velocity and gravity | Grades 3–5 · ages 9–11

**Objective:** Use a velocity variable to make a sprite fall faster and faster and land on the floor, and give it smooth sideways movement.

**Success criteria**

- I can use a velocity variable.
- I can code gravity that speeds up a fall.
- I can make a sprite land on the floor.

## Key vocabulary

- **velocity**: Speed with a direction (up is +, down is –).
- **gravity**: A pull that keeps changing the velocity downward.
- **frame**: One step of the game loop (about 30 per second).
- **floor**: The lowest place the sprite can stand.

## You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

## Timed lesson plan (45 min)

| Time | Stage | What happens |
|---|---|---|
| 5 min | **Warm-up (unplugged)** | Step-and-grow (5 min): a learner takes 1 step, then 2, then 3... while the class counts — that's acceleration. |
| 5 min | **Watch** | Play the lesson video with captions; pause at each heading and ask a question. |
| 8 min | **Build along** | Drop the frog: learners follow the steps. |
| 14 min | **Perform** | Code Lab: gravity, Follow the velocity |
| 9 min | **Probe** | Code Lab: walk and fall, activity |
| 4 min | **Wrap-up** | Exit ticket: What two blocks make gravity work every frame? |

## Unplugged warm-up

Step-and-grow (5 min): a learner takes 1 step, then 2, then 3... while the class counts — that's acceleration.

## Teaching the idea

A velocity variable stores how far the sprite moves per frame; gravity is a constant change to that velocity (acceleration). Landing clamps the position and zeroes the velocity. Separate polling loops handle horizontal input concurrently.

**Analogy:** Rolling down a hill: every second you roll a bit faster than the second before.

**Live demo:** Show “change y by -5” (constant fall) next to the vy version (accelerating fall). Ask which looks real.

## Questions to ask

| Ask | Listen for |
|---|---|
| What does vy store? | How far to move up or down each frame. |
| Why set vy to 0 on the floor? | So the sprite stops falling through the floor. |
| How do we make the Moon’s gravity? | Change vy by a smaller amount, e.g. -0.3. |

## Differentiation

**Support**
Provide the gravity loop; learners add the floor check.

**Stretch**
Add a terminal velocity: if vy < -15, set vy to -15.

## Common misconceptions

| Misconception | How to fix it |
|---|---|
| Gravity means changing y by a fixed number | Gravity changes the velocity; the velocity changes y. |
| The order of blocks in the loop doesn’t matter | Change vy, move, then check the floor. |

## Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

**Exit ticket**
What two blocks make gravity work every frame?

**Homework**
Drop a ball and a sheet of paper at home. Which speeds up more? Why might a game use different gravity?

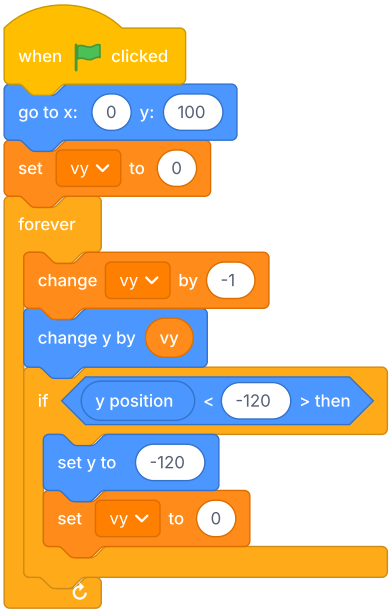
# Answer key — Kitabash activities

## 1. Code Lab: gravity (perform · code lab)

**Goals checked:** Moves by vy · Gravity changes vy · Lands on the floor

**One solution:**

Hero




## 2. Follow the velocity (perform · multiple choice)

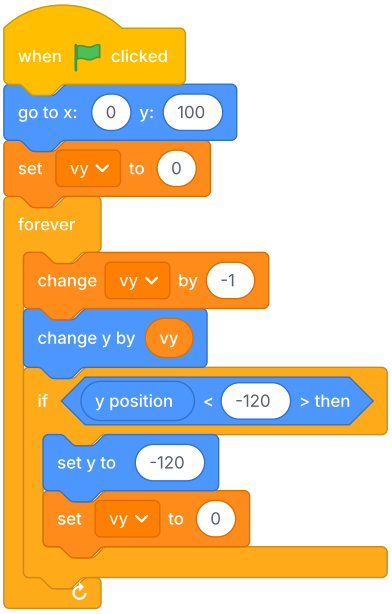
**Answer:** -5

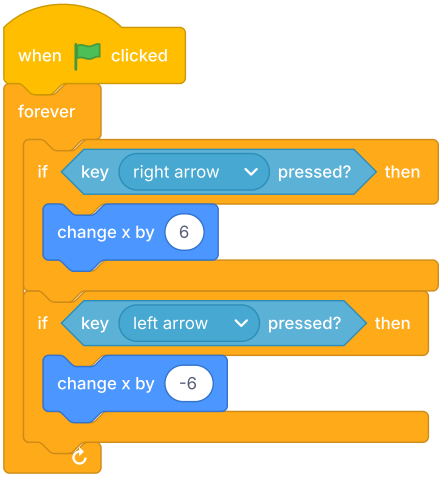
### 3. Code Lab: walk and fall (probe · code lab)

**Goals checked:** Holding → walks right · Holding ← walks left · Still lands

**One solution:**

Hero




### 4. Explain your thinking (probe · free response)

**Look for:** velocity, gravity, faster. **Example:** Gravity changes the velocity by -1 every frame, so the velocity gets more and more negative. The sprite moves by its velocity, so each frame it moves down further and falls faster.

## Answer key — worksheet

1. **Key words:** velocity = Speed with a direction (up is +, down is −). · gravity = A pull that keeps changing the velocity downward. · frame = One step of the game loop (about 30 per second). · floor = The lowest place the sprite can stand.
2. **Plan your code:** Goals: Moves by vy; Gravity changes vy; Lands on the floor.

3. **Follow the velocity:** A. -5 — -3 - 1 - 1 = -5: the sprite falls faster.
4. **Plan your code:** Goals: Holding → walks right; Holding ← walks left; Still lands.
5. **Explain your thinking:** Gravity changes the velocity by -1 every frame, so the velocity gets more and more negative. The sprite moves by its velocity, so each frame it moves down further and falls faster.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.